MR. ALLEN'S 8TH GRADE MATH

NC Check-In Benchmark Review

Version 2

Name _____ Date _____ Class _____

5 Test Tips

1. **Read it two times.**
   Read the question. Then read it again, slowly.

2. **Look at the example.**
   Each question has an example. Do the same steps.

3. **Cross out wrong answers.**
   If you know an answer is wrong, cross it out.

4. **Check your answer.**
   Put your answer back in. Does it work?

5. **Do not leave it blank.**
   Pick your best answer. Then go on.

Notes

NC.8.NS.1 Rational & Irrational Numbers

Rational = can be a fraction. The decimal **stops** or **repeats**.
Irrational = the decimal **never stops** and **never repeats**.

RATIONAL	IRRATIONAL
5 $\frac{1}{2}$ 0.25 $0.333\ldots$ $\sqrt{16}$	π $\sqrt{2}$ $\sqrt{7}$

Tip: Square root of a perfect square (1, 4, 9, 16, 25...) → rational. Not a perfect square → irrational.

NC.8.EE.1 Exponent Rules

An exponent tells how many times to multiply the base. $2^3 = 2 \times 2 \times 2 = 8$.

Multiply	add exponents	$2^3 \cdot 2^4 = 2^7$
Divide	subtract exponents	$5^6 \div 5^2 = 5^4$
Zero	always 1	$9^0 = 1$
Negative	flip it	$2^{-3} = \frac{1}{8}$

Tip: A negative exponent does NOT make a negative number.

NC.8.EE.7 Solving Equations

Do the **opposite** operation to both sides until x is alone.

$3x + 4 = 19$

$- 4 \rightarrow 3x = 15$

$\div 3 \rightarrow \mathbf{x = 5}$

Tip: Always check: put your answer back in. 3(5) + 4 = 19 ✓

NC.8.F.1 What Is a Function?

A **function** gives each x **only one** y.

x	1	2	3
y	4	4	6

Function ✓ — each x is used once. (A y can repeat.)

x	1	1	2
y	3	4	5

NOT a function × — x = 1 goes to two y's.

Tip: Graph? Use the vertical line test: touches only once → function.

NC.8.G.3 Moving Points

Use the rule. Then check on the grid.

Right / left	change **x**	$(2, 3)$ **right** $4 \rightarrow (6, 3)$
Up / down	change **y**	$(2, 3)$ **down** $1 \rightarrow (2, 2)$
Flip over x-axis	**y** changes sign	$(4, 2) \rightarrow (4, -2)$
Flip over y-axis	**x** changes sign	$(4, 2) \rightarrow (-4, 2)$

Tip: Right and up are plus. Left and down are minus.

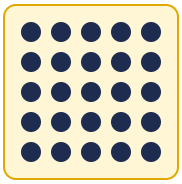
Questions

1 NC.8.NS.1 Rational or irrational?

EXAMPLE
$\sqrt{9} = 3$. 3 is a whole number. So $\sqrt{9}$ is **rational**.

Is $\sqrt{25}$ rational or irrational?

Ⓐ Rational Ⓑ Irrational



s make a 5 by 5 s

Show your work

2 NC.8.NS.1 Find the irrational number

EXAMPLE
$\pi = 3.14159...$ It never ends and never repeats. So π is **irrational**.

Which number is irrational?

Ⓐ 0.5 Ⓑ $\sqrt{7}$ Ⓒ $\frac{1}{4}$

1	4	9	16	25

Perfect squares: 1, 4, 9, 16, 25 ... Is 7 on the list?

Show your work

3 NC.8.EE.1 Multiply: add the exponents

EXAMPLE
$3^2 \cdot 3^5 = 3^{2+5} = \mathbf{3^7}$. Same base → **add** the exponents.

Simplify. $2^3 \cdot 2^4$

Ⓐ 2^7 Ⓑ 2^{12} Ⓒ 4^7



3 twos times 4 twos. Count all the twos.

Show your work

4 NC.8.EE.1 Negative exponent: flip it

EXAMPLE

$3^{-2} = \frac{1}{3^2} = \frac{1}{9}$. A negative exponent means **flip**.

What is 2^{-3}?

Ⓐ $\frac{1}{8}$ Ⓑ -8 Ⓒ -6



The negative sign goes away when you flip.

Show your work

5 NC.8.EE.7 Solve a two-step equation

EXAMPLE

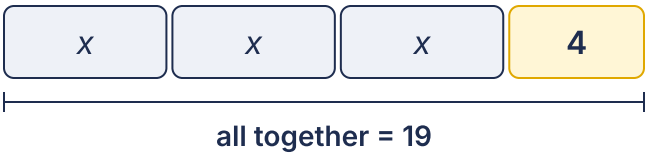
2x + 1 = 11

Subtract 1: 2x = 10

Divide by 2: **x = 5**

Solve. $3x + 4 = 19$

Ⓐ $x = 5$ Ⓑ $x = 7$ Ⓒ $x = 15$



Show your work

6 NC.8.EE.7 Variables on both sides

EXAMPLE

4x + 1 = 2x + 9

Subtract 2x: 2x + 1 = 9

Subtract 1: 2x = 8

Divide by 2: **x = 4**

Solve. $5x - 2 = 3x + 8$

Ⓐ $x = 5$ Ⓑ $x = 3$ Ⓒ $x = 10$

Show your work

7 NC.8.F.1 Is the table a function?

EXAMPLE

A function gives each x **only one** y.

x	1	2	3
y	4	5	6

Each x is used once. This **is** a function.

Is this table a function?

x	1	1	2
y	3	4	5

Ⓐ Yes Ⓑ No

Show your work

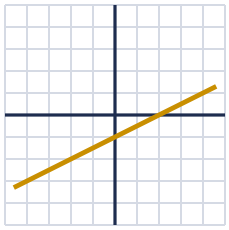
8 NC.8.F.1 Vertical line test

EXAMPLE

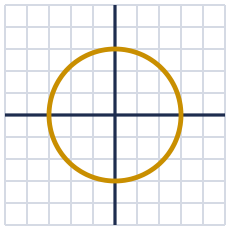
Slide a straight up-and-down line across the graph. If it touches the graph **only once** everywhere, it **is** a function.

Which graph is a **function**?

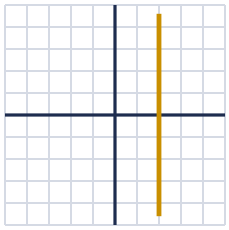
Ⓐ



Ⓑ



Ⓒ



Show your work

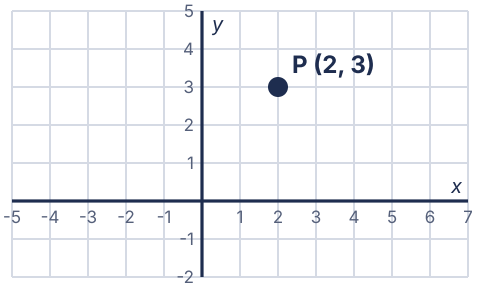
9 NC.8.G.3 Slide a point (translation)

EXAMPLE

Move (1, 1) right 2. Right means **add to x**. (1 + 2, 1) = **(3, 1)**.

Point P is at (2, 3). Move it **right 4**. Where does it land?

Ⓐ $(6, 3)$ Ⓑ $(2, 7)$ Ⓒ $(-2, 3)$



Show your work

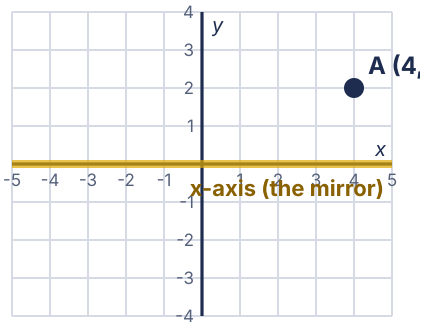
10 NC.8.G.3 Flip over the x-axis (reflection)

EXAMPLE

Reflect (3, 5) over the x-axis. The **y** changes sign. (3, 5) → **(3, −5)**.

Point A is at (4, 2). Reflect it over the x-axis. Where does it land?

Ⓐ (4, −2) Ⓑ (−4, 2) Ⓒ (2, 4)



Show your work

11 NC.8.EE.7 Score 5: Explain your thinking

EXAMPLE

2(x + 1) = 2x + 2. Both sides are the **same**, so every number works: **infinitely many solutions**.

x + 1 = x + 5. The x's match, but 1 is not 5: **no solution**.

Why does $3(x + 2) = 3x + 6$ have infinitely many solutions?

Ⓐ The x's are the same, so there is no solution.

Ⓑ After you distribute, both sides are the same: $3x + 6$.

Ⓒ Only $x = 0$ works.

Show your work